

6. Euclid's Algorithm

Definition:

Let m and n be natural numbers. The **greatest common divisor**, written as $\gcd(m, n)$ is the largest common divisor of m and n .

Divisors of 78	Divisors of 216
1	1
2	2
3	3
6	4
13	6
26	9
39	12
78	18
	24
	36
	54
	72
	108
	216

So, $\gcd(78, 216) = 6$.

This approach is not very efficient. Euclid's Algorithm, written about 2500 years ago, helps us find the greatest common divisor of two numbers in a more efficient way.

Euclid's Algorithm:

1. Let $n > m$
2. Divide n by m and calculate the remainder r . Note that $0 \leq r < n$.
Then write $n = km + r$.

3. Repeat step 2, but replace n with m , and m with r , i.e., write

$$m = k'r + r'$$

Repeat step 2 again, this time replace m with r and r with r'

$$r = k''r' + r''$$

This is repeated until we obtain a remainder zero.

4. When the remainder is zero, the $\gcd(m, n)$ is the last non-zero remainder obtained.

Example 1:

Find $\gcd(78, 216)$.

Solution:

$$216 = 2 \times 78 + 60$$

$$78 = 1 \times 60 + 18$$

$$60 = 3 \times 18 + 6$$

$$18 = 3 \times 6 + 0$$

$$\therefore \gcd(78, 216) = 6$$

Example 2:

Find $\gcd(1568, 890)$

Solution:

$$1568 = 1 \times 890 + 678$$

$$890 = 1 \times 678 + 212$$

$$678 = 3 \times 212 + 42$$

$$212 = 5 \times 42 + 2$$

$$42 = 21 \times 2 + 0$$

$$\therefore \gcd(1568, 890) = 2$$

Writing the $\gcd(m, n)$ as a linear combination of m and n :

If m and n are any two natural numbers, then we can always find integers s and t such that

$$\gcd(m, n) = sm + tn$$

To find s and t we use the steps carried out in the Euclid algorithm, but backwards.

Example 3:

Find s and t such that $\gcd(78, 216) = 78s + 216t$

Solution:

$$216 = 2 \times 78 + 60 \quad \Rightarrow 60 = 216 - 2 \times 78$$

$$78 = 1 \times 60 + 18 \quad \Rightarrow 18 = 78 - 1 \times 60$$

$$60 = 3 \times 18 + 6 \quad \Rightarrow 6 = 60 - 3 \times 18$$

$$18 = 3 \times 6 + 0$$

Hence:

$$\begin{aligned} 6 &= 60 - 3 \times 18 = 60 - 3(78 - 60) \\ &= 4 \times 60 - 3 \times 78 \\ &= 4(216 - 2 \times 78) - 3 \times 78 \\ &= -11 \times 78 + 4 \times 216 \end{aligned}$$

Hence, $s = -11$ and $t = 4$

Example 4:

Find s and t such that $\gcd(1568, 890) = 1568s + 890t$.

Solution:

$$1568 = 1 \times 890 + 678 \quad \Rightarrow \quad 678 = 1568 - 1 \times 890$$

$$890 = 1 \times 678 + 212 \quad \Rightarrow \quad 212 = 890 - 1 \times 678$$

$$678 = 3 \times 212 + 42 \quad \Rightarrow \quad 42 = 678 - 3 \times 212$$

$$212 = 5 \times 42 + 2 \quad \Rightarrow \quad 2 = 212 - 5 \times 42$$

$$42 = 21 \times 2 + 0$$

Hence,

$$\begin{aligned} 2 &= 212 - 5 \times 42 = 212 - 5(678 - 3 \times 212) \\ &= 16 \times 212 - 5 \times 678 \\ &= 16(890 - 1 \times 678) - 5 \times 678 \\ &= 16 \times 890 - 21 \times 678 \\ &= 16 \times 890 - 21(1568 - 1 \times 890) \\ &= -21 \times 1568 + 37 \times 890 \end{aligned}$$

Hence, $s = -21$ and $t = 37$.